

Arthur Allilaire

📍 London

✉ arthur.allilaire@gmail.com

☎ 07752 715719

🌐 Arthur-Allilaire

🔗 ArthurAllilaire

Profile

- **Founder in Residence @ The Bridge (EF)** — 8-week residency in San Francisco.
- **Imperial CS BEng (82%)** — awarded the **Aspect Capital Project Prize** for my project using RL to speed up MRI.
- **Internships** (AI systems & data integrations): **Palantir** SWE; **Kayrros** — Junior ML Engineer (RAG); **Clear** (YC startup) — Vision ML. Plus **2 years part-time as a data developer @ RLT** during Imperial.
- **French & British citizenship.**

Experience

The Bridge (Entrepreneurs First), Founder in Residence

San Francisco | Jul 2026 – present

- 8-week [residency](#) in San Francisco building a company from scratch.

Palantir, Software Engineer Intern, Data Federation Team

London | Jul 2025 – Sep 2025

Improved performance and interoperability of Palantir's Foundry platform with external data systems.

- Optimized batch registration of virtual tables (external datasets), reducing latency by up to **40%** for large requests.
- Enabled wholistic data governance by **integrating external data lineage** from **Databricks** and **Snowflake**.
- Full-stack development: Micro-service infrastructure (Java + [Conjure](#)), React front-end & Kubernetes deployment. **Received a return offer.**

River Learning Trust (RLT), Data Developer (Part-Time)

Oxford (Remote) | Oct 2024 – Jul 2026

Sole data engineer for a 30+ school trust, owning the data infrastructure end to end. Worked part-time (10-15 hrs/week) alongside studies at Imperial, migrating the trust-wide architecture from **spreadsheets** to **Google BigQuery**.

- Migrated the annual stakeholder survey (**30k responses**) to a web portal backed by BigQuery + Looker Studio — **now sold to 20+ external schools**. Optimised pipelines to **cut cloud costs by 4x**.
- Built automated attendance pipelines covering **16,000 pupils**, reducing dashboard load times from **30s to under 5s**.
- Acted as **on-call engineer** for production failures; impact led to the creation of a full-time role. Work recognised in RLT's receipt of **Google's Excellence in Education Award**.

Kayrros, Junior Machine Learning Engineer

Paris | Apr 2024 – Jun 2024

Kayrros is a leading climate and energy data analytics scale-up (\$18 million revenue, 130 employees)

- Helped design and implement the architecture for [KayrrosAI](#), a **Claude-powered RAG** that uses text embeddings, and generated SQL queries to retrieve and analyse Kayrros data. (Hosted on AWS)
- Improved test-dataset performance **from 50% to 90%** using prompt engineering, retry loops and pre-processing layers. Built a rigorous benchmark dataset with subject experts, tested via a custom LLM evaluation architecture.

Clear (YC-backed skincare app), Vision ML Intern

London | Jul 2022 – Aug 2022

- Built a **Tensorflow** model to measure acne severity (deployed with Kubernetes & Flask), optimised via image pre-processing.

Education

Imperial College London, Computing, BEng

London | Oct 2023 – Jun 2026

- **Grade: 82% (First)**, awarded **The Aspect Capital Project Prize**.
- **Final year project (88%)**: [Reinforcement Learning for Adaptive Quantitative MRI in a Validated Digital Phantom](#) — research collaboration with the Institute of Cancer Research. Submitting to **ISMRM 2027**.
- Studied Functional Programming (Haskell & Scala), OOP (Kotlin, Java), Databases (SQL) and Imperative Programming (C)
- **Group projects**: [SimPort](#), a Kotlin library and DSL for simulating shipping ports; an operating system in C ([Pintos](#)); a [WACC](#) compiler in Scala targeting ARM 64 and X86 (Pintos + WACC grade: 88%).

City of London School, A-levels & GCSEs

London | Sep 2015 – Jul 2023

- **A-levels**: Maths (94%), Further Maths, Physics (92%) and Economics (4 A*); **GCSE**: 11× Grade 9
- **Head of Website, iGEM**: led 3 designers building the [React site](#) for our **Gold-winning** synthetic biology project.
- Self-taught uni courses: *Intro to Python* (MIT), *Intro to ML* (Stanford), *CS50 Web Dev and AI* (Harvard) [Code](#)

Volunteering

City of London School, Chair of Charity

London | Sep 2021 – Jul 2022

- Led the annual charity appeal, raising **£65,000** for Papyrus (**up 45% YoY**), coordinating a 16-member student committee to organise fundraising events.

Bow Foodbank, Volunteer

London | Oct 2016 – Oct 2024

- Accumulated over **250 volunteer hours**, working in teams to receive, organise and stock food deliveries.